Exponential negligibility of the far-phase region

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Abstract

Astra #11 rank 5. The proof of thm:strataempiricalexpansion splits the partition function into the region $K < \varepsilon$ (resolution and Taylor tree) and the far region $K \geq \varepsilon$, where the AM–GM inequality $\beta\sqrt{nK}\,\psi \leq \beta(nK+\psi^2)/2$ makes the integrand exponentially small. We formalise the far-region step on an arbitrary measurable set E of a measure space: if $K \geq \varepsilon \geq 0$ and $|\psi| \leq M$ on E and η is integrable on E, then

$$\Big|\int_E \eta\, e^{-\beta nK+\beta\sqrt{nK}\psi}\, d\mu\Big| \leq \Big(\int_E |\eta|\, d\mu\Big) e^{-\beta n\varepsilon/2+\beta M^2/2}$$

(`farPhase_bound`), with exponential decay in n for fixed M (`farPhase_tendsto_zero`). Zero `sorry`/`axiom`.

1 The things formalised

```
theorem farPhase_exponent_le ( n  M K ) (h : 0 < ) (hn : 0  n) (h : 0  )
    (hK :   K) (h : | |  M) :
    - * n * K +  * Real.sqrt (n * K) *   - * n *  / 2 +  * M ^ 2 / 2
theorem farPhase_bound ( : Measure W) (E : Set W) (hE : MeasurableSet E) ( n  M)
    (h  hn h ) ( K  : W →  ) (hK :  w  E,   K w) (h :  w  E, | w|  M)
    (h : IntegrableOn  E ) :
    | w in E,  w * exp (- * n * K w +  * Real.sqrt (n * K w) *  w) |
       ( w in E, | w|  ) * exp (- * n *  / 2 +  * M ^ 2 / 2)
theorem farPhase_tendsto_zero (  M I) (h : 0 < ) (h : 0 < ) :
    Tendsto (fun n :  => I * exp (- * n *  / 2 +  * M ^ 2 / 2)) atTop ( 0)
```

2 Mathematics

For $nK \geq 0$, $2\sqrt{nK}\,|\psi| \leq nK + \psi^2$, so $-\beta nK + \beta\sqrt{nK}\psi \leq -\beta nK/2 + \beta\psi^2/2 \leq -\beta n\varepsilon/2 + \beta M^2/2$. The integrand is therefore dominated pointwise on E by $|\eta|$ times a constant, and the integral bound follows from `norm_integral_le_of_norm_le` on the restricted measure. The paper's formulation uses $\sup_{K\geq\varepsilon}|\psi_n|$; we take an explicit bound M, from which the supremum version is immediate. Astra #11 warned that "outside a small box" is not a valid region for the monomial phase $u_1^{2k_1}u_2^{2k_2}$ (it approaches the axes); the region must be $\{K \geq \varepsilon\}$, which is how the theorem is stated.

3 Paper-facing meaning

This is the $Z_n^{(2)}$ step of thm:strataempiricalexpansion: with $M_n^2 = \sup_{K\geq\varepsilon}\psi_n^2$ bounded in probability, $Z_n^{(2)} = O_p(e^{-\beta\varepsilon n/2})$, hence $o_p(e^{-cn})$ for every $c < \beta\varepsilon/2$ (the paper writes $o_p(e^{-\varepsilon' n})$ for some

$\varepsilon' > 0$, which is what this gives). Together with units 145–150 the chart-level analytic content of §4.3 for $d = 2$ is now formalised; what remains is probabilistic input (the functional CLT, tightness) and the resolution/chart assembly. Astra #11 ranks 1–5 are complete.

4 Lean notes

- `two_mul_le_add_sq a b : 2 * a * b   a^2 + b^2` — the statement must be written left-associated (`2 * √(nK) * | |`), not `2 * (√(nK) * | |)`.
- The exponent inequality is one `nlinarith` call with the AM–GM fact, `Real.sq_sqrt`, $\psi^2 \leq M^2$ and $n\varepsilon \leq nK$ supplied (after `rw [sq_abs]`).
- `ae_restrict_of_forall_mem hE` converts a pointwise bound on E to an a.e. bound for `.restrict E`; `integral_mul_const` finishes.
- Exponential decay: `tendsto_id.const_mul_atTop_of_neg`, then `tendsto_atBot_add_const_right`, then `Real.tendsto_exp_atBot.comp` (there is no `Tendsto.atBot_add_const`).